QISKIT FALL FEST 2026 / UNIVERSITY OF STUTTGART

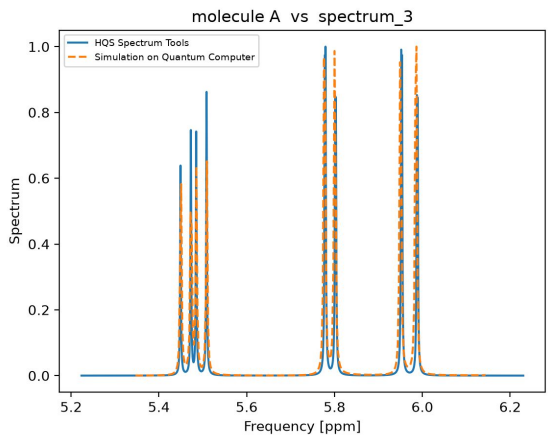
Matching Molecules to NMR Spectra with Qiskit

Quantum simulation of liquid-state ^{1}H NMR

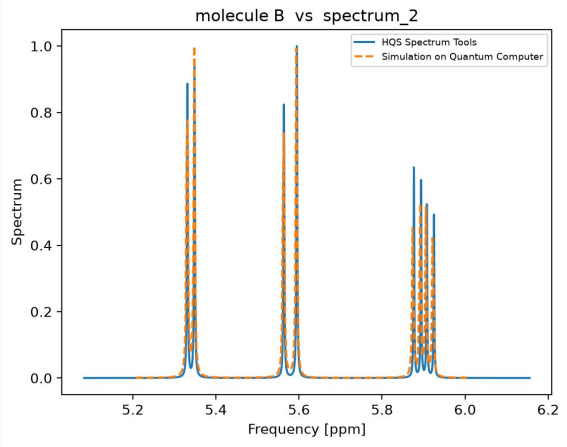
Three spin Hamiltonians. Three reference spectra.
One molecular matching task.

Matching Molecules to Reference Spectra

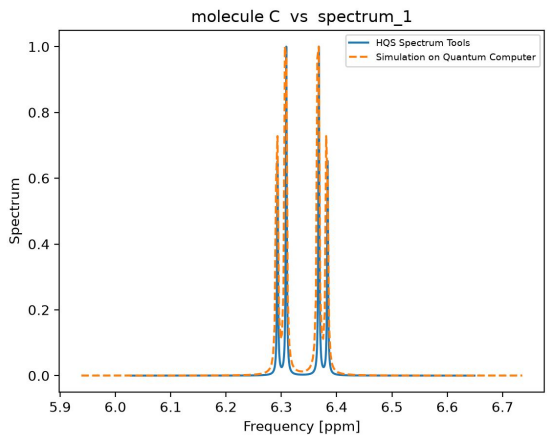
Molecule A -> Spectrum 3



Molecule B -> Spectrum 2



Molecule C -> Spectrum 1



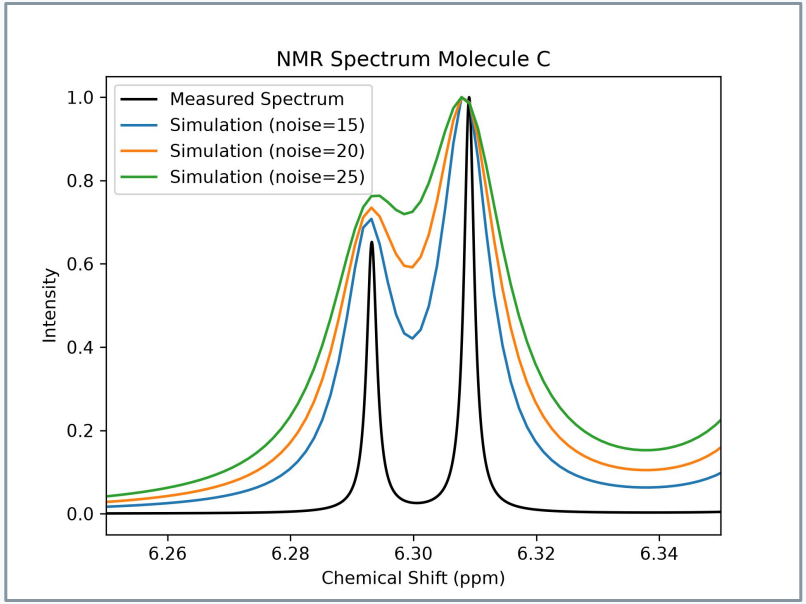
Influence of Noise Strength

$$M(t) = e^{-\gamma t}$$

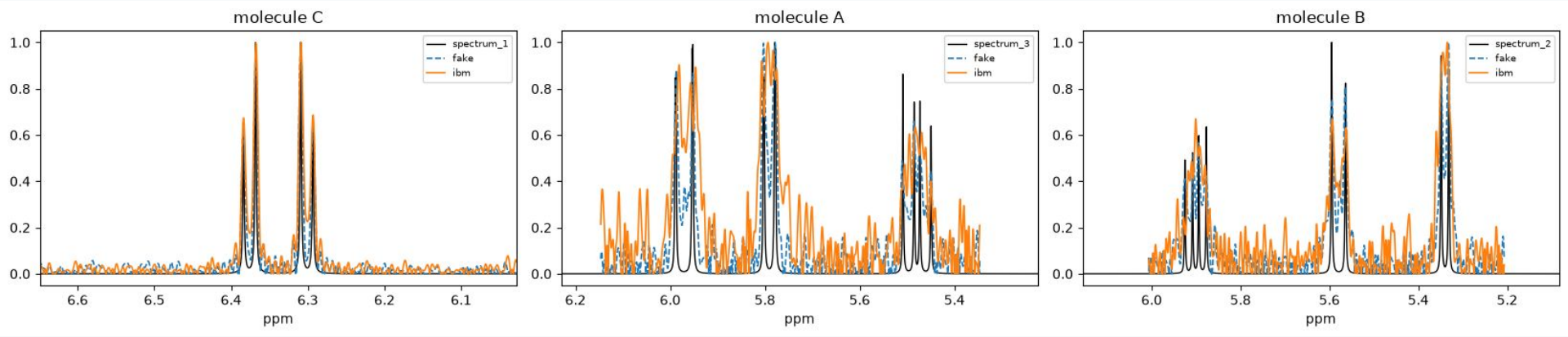
Noise strength γ controls decay of magnetization over time

Molecule	γ max (s^{-1})	Smallest |J| (Hz)	Smallest visible peak distance (Hz)
A	≈ 17.0	0.9	2.79
B	≈ 20.0	0.5	2.86
C	≈ 20.0	7.92	2.53

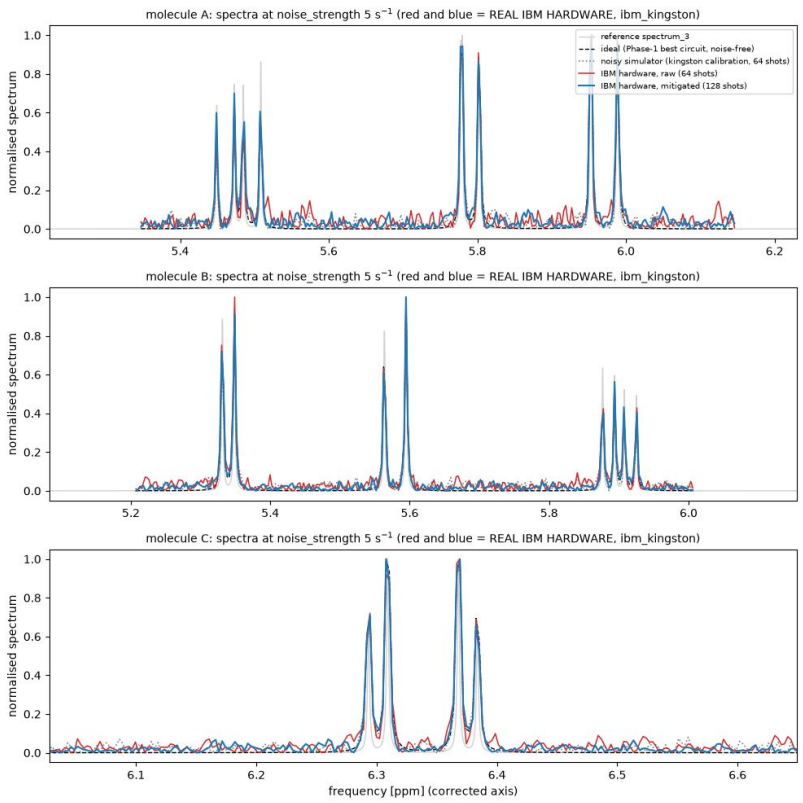
γ max is the largest value the noise parameter can take, such that the peaks can be resolved.
Peaks count as resolved if the dip between two peaks is at least 17 % of the smaller peak
(corresponds to: separation = linewidth for a Laurentzian fit)



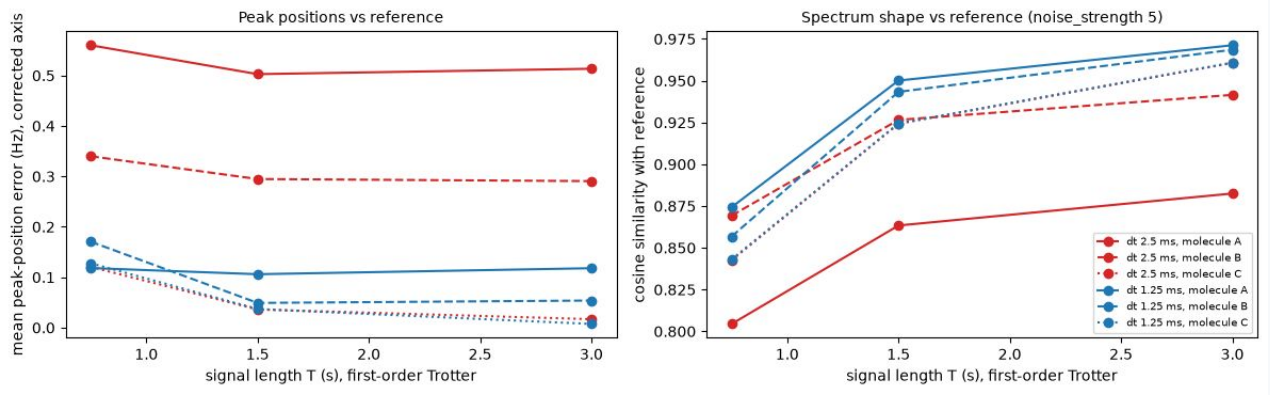
Simulations on Quantum Computer

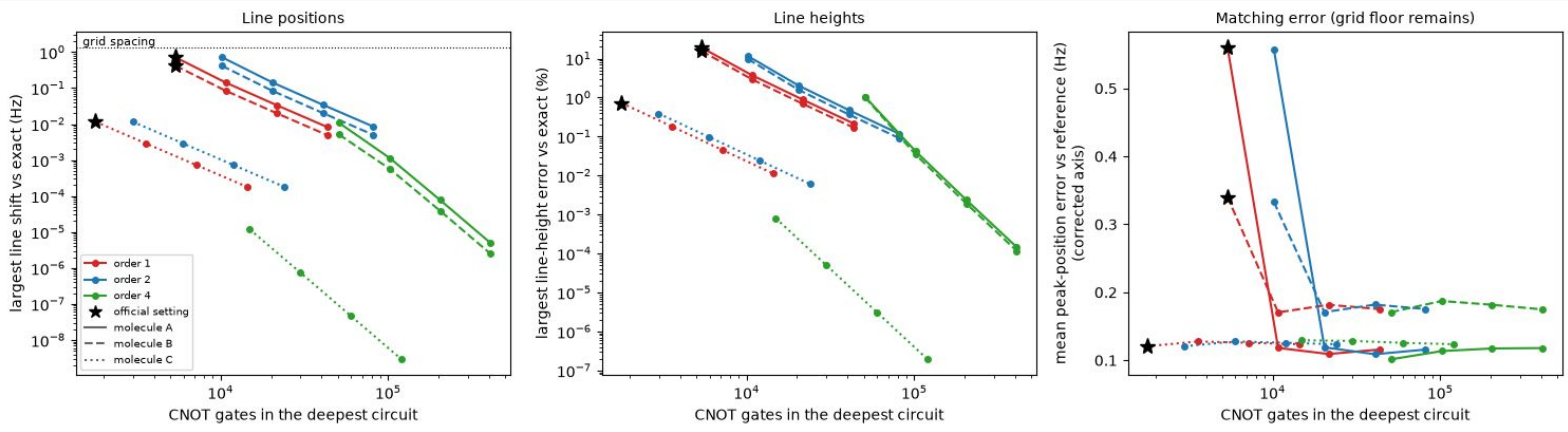


Simulations on Quantum Computer



Extending Signal Length





Conclusions and Limitations

What the results show

MATCHING

A → 3 B → 2 C → 1

NOISE STRENGTH

- Largest possible noise strength of the simulation increases with an increasing coupling strength

What the model leaves out

- Statevector simulation only; no IBM QPU run
- No quantum error correction
- Frequency resolution is limited due to the trotterization error and the limited simulation time of the magnetization